COURSE CODE/TITLE: CPE 010-CPE21S1 - Data Structures and Algorithms	**SECTION: CPE21S1**
DATE PERFORMED: July 09 2026	**DATE SUBMITTED: Jul 13, 2026**
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ACTIVITY NO. 2 **Hands-on Activity 1.1 Basic C++ Programming**

1. PROCEDURE OUTPUT

ILO A:

TASK 1: Create a function that will take two numbers and display the sum.

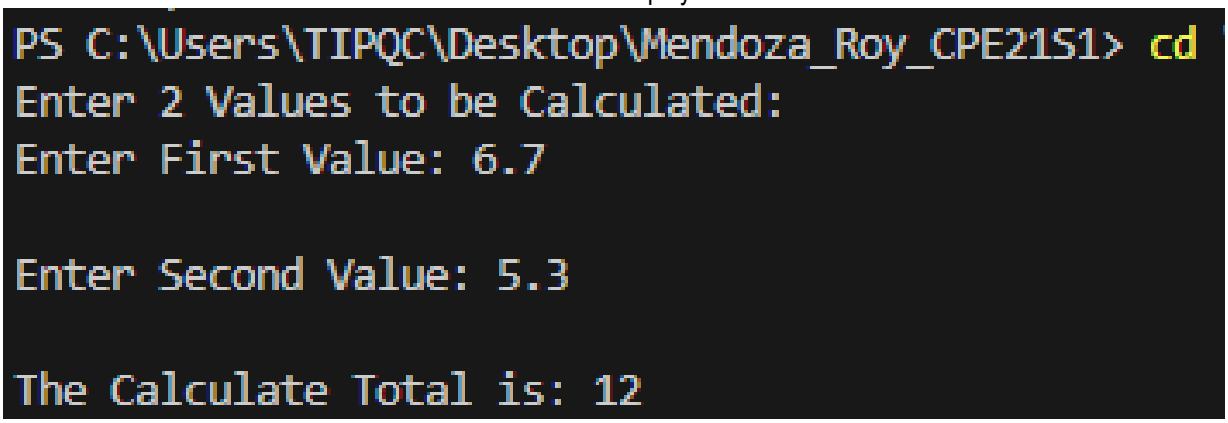


Figure 1: Output of the Code of Task 1

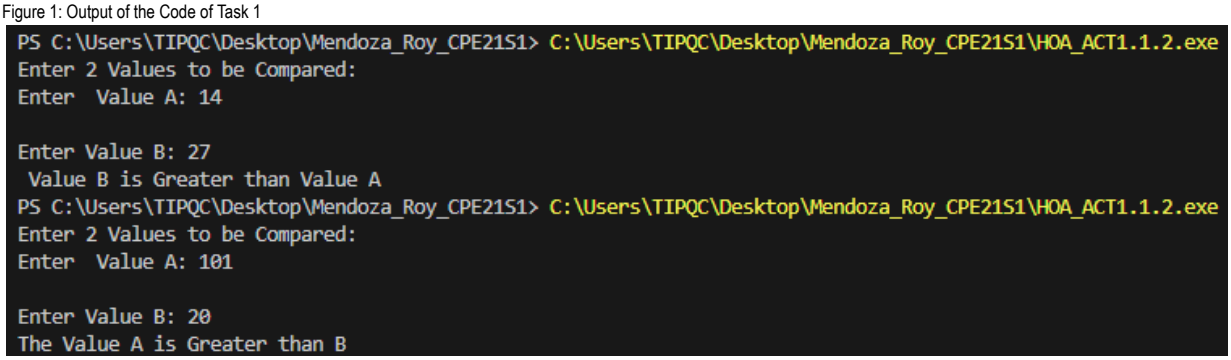


Figure 2: Output of the Code of Task 2

```
PS C:\Users\TIPQC\Desktop\Mendoza_Roy_CPE21S1> C:\Users\TIPQC\Desktop\Mendoza_Roy_CPE21S1\HOA_ACT1.1.3.exe
Enter 2 Values to be Calculated:
Enter First Value: 1

Enter Second Value: 1
For AND: 1
For OR: 1
For NOT of Value 1: 0
For NOT of Value 2: 0
PS C:\Users\TIPQC\Desktop\Mendoza_Roy_CPE21S1> C:\Users\TIPQC\Desktop\Mendoza_Roy_CPE21S1\HOA_ACT1.1.3.exe
Enter 2 Values to be Calculated:
Enter First Value: 0

Enter Second Value: 0
For AND: 0
For OR: 0
For NOT of Value 1: 1
For NOT of Value 2: 1
PS C:\Users\TIPQC\Desktop\Mendoza_Roy_CPE21S1> 1
1
PS C:\Users\TIPQC\Desktop\Mendoza_Roy_CPE21S1> 0
0
PS C:\Users\TIPQC\Desktop\Mendoza_Roy_CPE21S1> C:\Users\TIPQC\Desktop\Mendoza_Roy_CPE21S1\HOA_ACT1.1.3.exe
Enter 2 Values to be Calculated:
Enter First Value: 1

Enter Second Value: 0
For AND: 0
For OR: 1
For NOT of Value 1: 0
For NOT of Value 2: 1
PS C:\Users\TIPQC\Desktop\Mendoza_Roy_CPE21S1> C:\Users\TIPQC\Desktop\Mendoza_Roy_CPE21S1\HOA_ACT1.1.3.exe
Enter 2 Values to be Calculated:
Enter First Value: 0

Enter Second Value: 1
For AND: 0
For OR: 1
For NOT of Value 1: 1
For NOT of Value 2: 0
```

Figure 3: Output of the Code of Task 3

ILO B:

Sections	Answer
Header File Declaration Section	#include <iostream>

Global Declaration Section	`double sumTOTnum(double first, double second);`
	`;bool great(int first, int second);`
	`;bool lgxop(bool one, bool two);`

Class Declaration and	None

Method Definition Section	
Main Function	

```
int main(){ //driver code

   double first, second;

   std::cout<<"Enter 2 Values to be
Calculated: "<<std::endl;
   std::cout<<"Enter First Value: ";
   std::cin >> first;
   std::cout <<std::endl<<"Enter
Second Value: ";
   std::cin >> second;
   std::cout <<std::endl<<"The
Calculate Total is: "
<<sumTOTnum(first,
second)<<std::endl;

   return 0;
}
int main(){ //driver code

   int first, second;
   bool result;

  std::cout<<"Enter 2 Values to be
Compared: "<<std::endl;
  std::cout<<"Enter  Value A: ";
  std::cin >> first;
  std::cout <<std::endl<<"Enter
Value B: ";
  std::cin >> second;
  result = first>second;
  if(result == true){
       std::cout<<"The Value A is
Greater than B" ;
   } else{
       std::cout<<" Value B is
Greater than Value A" ;
   }
   return 0;
}
int main(){ //driver code
   bool one, two;

   std::cout<<"Enter 2 Values to be
Calculated: "<<std::endl;
   std::cout<<"Enter First Value: ";
   std::cin >> one;
   std::cout <<std::endl<<"Enter
Second Value: ";
   std::cin >> two;
```

	`lgxop(one, two);`
	`return 0;`
	`}`

Method Definition	`double sumTOTnum(double first,double second){`
	`double tot = first+second;`
	`bool great(int first, int second){`
	`bool result = first>second;`
	`bool lgxop(bool one, bool two){`
	`std::cout<<"For AND: "<<(one&&two)<< std::endl;`
	`std::cout<<"For OR: "<<(one||two)<<std::endl;`
	`std::cout<<"For NOT of Value 1: "<< (!one)<<std::endl;`
	`std::cout<<"For NOT of Value 2: "<< (!two)<<std::endl;`

Table 1-1. C++ Structure Code for Answer

>> The constructor was able to identify that the angles 40, 30 and 110 are valid triangles. >> This part of the activity shows the classes public and private data. >> also, according to the Google, a triangle specifically or always adds up to exactly 180, however the code on the manual is using the <= which means 180 and below would result to a triangle.

Table 1-2. ILO B output observations and comments

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary Activity
ILO C: Solve Different Problems using the C++ Programming Language
The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.
1. Create a C++ program to swap the two numbers in different variables.

```
C:\Users\royga\CLionProjects\untitled1\ILOC-1.exe
Enter value for the First No.:12

Enter value for Second No.:52

BEFORE SWAPPING: THESE ARE THE VALUES OF THE FIRST AND SECOND VARIABLE:
FIRST VALUE: 12
SECOND VALUE: 52
FIRST VALUE: 52
SECOND VALUE: 12

Process finished with exit code 0
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```
C:\Users\royga\CLionProjects\untitled1\ILOC2.exe
Enter the Temperature in Kelvin:120

 The Temperature of 120k in Fahrenheit is: -243.67

Process finished with exit code 0
```

3. Create a C++ program that has a function that will calculate the distance between two points.

```
C:\Users\royga\CLionProjects\untitled1\ILOC-3.exe
Enter the Coordinates of the First Point(x1,y1):
x1:20

y1:30

Enter the Coordinates of the Second Point(x1,y1):
x2:15

y2:25


The Distance between two Points is: 7.07107
Process finished with exit code 0
```

4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle

```
C:\Users\royga\CLionProjects\untitled1\ILOC-4.exe
Enter Angle A:
30

Enter Angle B:
50

Enter Angle C:
100

Enter Side A:
5

Enter Side B:
6

Enter Side C:
7

=====Menu Selection=====:

1. Compute Area
2. Compute Perimeter
3. Determine Triangle Type
Enter Choice:1

Area of Triangle: 14.6969

Process finished with exit code 0
```

b. A function to compute for the perimeter of a triangle

```
C:\Users\royga\CLionProjects\untitled1\ILOC-4.exe
Enter Angle A:
30

Enter Angle B:
50

Enter Angle C:
100

Enter Side A:
5

Enter Side B:
6

Enter Side C:
7

=====Menu Selection=====:

1. Compute Area
2. Compute Perimeter
3. Determine Triangle Type
Enter Choice:2

Perimeter of Triangle: 18

Process finished with exit code 0
```

c. A function that determines whether the triangle is acute-angled, obtuse-angled or ‘others.’

```
C:\Users\royga\CLionProjects\untitled1\ILOC-4.exe
Enter Angle A:
30

Enter Angle B:
50

Enter Angle C:
100

Enter Side A:
5

Enter Side B:
6

Enter Side C:
7

=====Menu Selection=====:

1. Compute Area
2. Compute Perimeter
3. Determine Triangle Type
Enter Choice:3

The triangle is Obtuse-Angled.

Process finished with exit code 0
```

8. Conclusion

Provide the following:

1. **Summary of lessons learned:** The lessons that i've learned that are new, is the proper creation of the classes, also the proper citing? Or the separation of the declaration, main driver and the methods

2. **Analysis of the procedure:** for the procedures of this activity, for ILO A, this part is quite easy(feel new since we are not using namespace std anymore), for ILO B, since it is mostly just copy and pasting and analyzing, there is not much problem, but I did need to search from Google on how to get or determine how it is a triangle or not.

3. **Analysis of the supplementary activity:** On this part, It became hard for me, Specifically, the 3rd/c part, since I do not know how to determine the type of triangle. I did search on Google how it is determined but I still do not quite grasp it correctly. I did also search for an example of this part(entirely basically from codescracker on how to do the calculations/formula(the heron's formula was basically used here)). Also with the use of the CodesCracker, I was able to switch from making everything be outputted to a switch design for the selection of the process.
4. **Concluding statement / Feedback: How well did you think you did in this activity? What are your areas for improvement?**
   For all that I've done here and the time spent, I would give myself a 6 for time consumption that it took me to make all of this(basically the class time and July 13 from 7pm to 10:38pm), a 4 for the formula, 8 for ai use(basically just search for a shortcut), and a 7 for the knowledge in making the code/structure ((all are over 10)). I need to get the formulas (impossible) and make my thoughts more clear and follow a steady way from the start.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google AI overview**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **To search any method to swap variables.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **It showed me that there is a function called swap() which swaps the variable's element. It did not replace the original for it only shows that there is a function like that.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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